Networks and Complexity

Solution 20-2

This is an example solution from the forthcoming book Networks and Complexity. Find more exercises at https://github.com/NC-Book/NCB

Ex 20.2: Zero power [Lvl. 1]
Consider $\sum \lambda_n{}^k$, where λ_n were the eigenvalues of the adjacency matrix $\mathbf{A}$. This sum reveals interesting network properties for several values of k. Is this also true for $k = 0$?

Solution
Arguably this is the number of nodes N. We can write

$$\sum_{n=1}^{N} \lambda_n{}^0 = \sum_n 1 = N. \tag{1}$$

We have to be a little bit careful here, because the eigenvalues may be zero. The meaning of the expression 0^0 depends on the context. In calculus we would usually discover its meaning by a careful consideration of limits. But our eigenvalues live in the world of algebra where we can't do this, so in the present context it is acceptable to define $0^0 = 1$.